Outliers of Greatness

A Statistical Analysis of Dominance in Competitive Domains.

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Abstract

This paper investigates whether statistical tools such as standard deviation and z-scores can be used to compare exceptional performance across sports, esports and other competitive fields. By analyzing selected athletes and players relative to their competitive fields, this study aims to explore whether greatness can be measured through statistical deviation from the average.

Preface

This paper began as a project for Grade 12 Data Management (MDM4U) in Ontario. Although it was originally created for a high school course, I wanted to approach it as a research-style statistical investigation by collecting real-world data, applying mathematical tools, and presenting the results in a formal format.

After processing said data, we will create connections between different competitive domains and find the links between such fields. This allows us to answer a question in pop-culture with a more objective framing, whether or not Geometry dash is harder than Osu!

This is my first attempt at writing a mathematical paper using LaTeX, and it represents an early step in learning how to communicate quantitative ideas in a more rigorous and professional way.

1 Introduction

The ontological construction of greatness in competitive domains necessitates a rigorous epistemological framework.

In other words... debates of who the "greatest" is in competitive fields is often subjective. This research paper investigates whether statistical measures such as standard deviation and z-scores can provide a more objective way to compare dominance across different competitive fields.

Please note, the analysis and interpretation are all my own, so statistics and conclusions may be skewed by false assumptions. It is difficult and possibly impossible to objectively construct a measurable system for how "great" somebody is in a specific domain due to all the factors.

I will however do my best to consider all factors and weigh them in the most objective and comprehensive system I can create. My methods of calculation and factors are explained in Statistical Backgrounds.

2 Research Question

Main question:

Can statistical analysis methods such as z-scores be used to compare dominance between competitive domains?

Sub-topics:

- What is the hardest game?

- Can different domains be compared fairly using the same statistical methods?

- Which competitive domain has the largest statistical outlier?

3 Statistical Background

Here is a brief, non-comprehensive list of the statistical methods I will use to process raw data in this paper.

These are the tools for quantifying data, not the methods used for analysis.

3.1 Statistical Vectors

Since each competitor may have multiple statistics, their performance can be represented as a vector. For example, a competitor's standardized statistics can be written as:

$$\vec{z} = \begin{bmatrix} z_1 \\ z_2 \\ z_3 \\ \vdots \\ z_n \end{bmatrix}$$

where each z_i represents the z-score of one statistic.

A weighted composite score can then be interpreted as a dot product between a weight vector and a standardized statistics vector:

$$S = \vec{w} \cdot \vec{z}$$

where $\vec{w}$ is the weight vector and $\vec{z}$ is the standardized performance vector.

3.2 Mean

For ungrouped data, the mean is calculated as:

$$\mu = \frac{\sum x_i}{n}$$

For data with frequencies, the mean is calculated as:

$$\mu = \frac{\sum f_i x_i}{\sum f_i}$$

3.3 Standard Deviation

Standard deviation measures how spread out the values in a dataset are. A low standard deviation means the values are close to the mean, while a high standard deviation means the values are more spread out.

For population data, standard deviation is calculated as:

$$\sigma = \sqrt{\frac{\sum (x_i - \mu)^2}{n}}$$

For sample data, standard deviation is calculated as:

$$s = \sqrt{\frac{\sum (x_i - \bar{x})^2}{n-1}}$$

For data with frequencies, the population standard deviation is:

$$\sigma = \sqrt{\frac{\sum f_i (x_i - \mu)^2}{\sum f_i}}$$

3.4 Z-Score

A z-score measures how many standard deviations a value is above or below the mean. It is useful for comparing values from different datasets because it standardizes them onto the same scale.

$$z = \frac{x - \mu}{\sigma}$$

where x is the observed value, μ is the mean, and σ is the standard deviation.

3.5 Weighted Composite Score

A weighted composite score combines multiple standardized statistics into one overall score. In this paper, each statistic should first be converted into a z-score before being weighted, so that different types of statistics can be compared on the same scale.

$$S = w_1 z_1 + w_2 z_2 + w_3 z_3 + \cdots + w_n z_n$$

where S is the final composite score, z_i represents the z-score of each statistic, and w_i represents the weight assigned to that statistic.

4 Methodology

4.1 Domain Selection

The sample domains were chosen to represent a variety of competitive fields, including traditional sports, esports, rhythm games, intellectual competitions, and popularity-based platforms.

Category	Sample Domains
Traditional Sports	Soccer, Swimming, MMA, Weightlifting
Esports and Games	Valorant, Tetris, osu!, Geometry Dash, Aiming
Intellectual Competitions	Chess, Olympiads, MCTiers, Rubik's Cubing
Digital Popularity	Social media popularity

Table 1: Sample competitive domains considered for analysis

4.2 Metrics Chosen

Because each competitive domain measures performance differently, the raw statistics cannot be compared directly. To make comparison possible, each domain will first be evaluated using a weighted composite score. This score combines several domain-specific statistics into one standardized value.

For each domain, I will consider three levels of comparison:

1. the average professional competitor,
2. the average of the top ten competitors,
3. the highest ranked competitor compared to the second highest ranked competitor.

The average professional competitor provides a baseline for the field. The top ten average shows how exceptional the best competitor is among elite competitors. The gap between the first and second ranked competitors measures whether the best competitor is clearly separated from their closest rival.

4.3 Comparison Methods

After calculating the score for each competitor, I will compare the best competitor in each domain using several measurements. These comparisons are meant to measure not only who has the highest score, but how far that competitor is separated from the rest of the field.

First, I will calculate the best competitor's standardized separation from the average competitor:

$$\alpha = z_{\text{avg gap}} = \frac{S_{\text{best}} - \mu_{\text{field}}}{\sigma_{\text{field}}}$$

where S_{best} is the best competitor's score, μ_{field} is the mean score of the comparison field, and σ_{field} is the standard deviation of the comparison field. This measures how many standard deviations the best competitor is above the average competitor.

Second, I will calculate the best competitor's separation from the top ten competitors:

$$\beta = z_{\text{elite}} = \frac{S_{\text{best}} - \mu_{\text{top 10}}}{\sigma_{\text{top 10}}}$$

where $\mu_{\text{top 10}}$ is the mean score of the top ten competitors and $\sigma_{\text{top 10}}$ is the standard deviation of the top ten competitors. This measures whether the best competitor is still an outlier among elite competitors.

Third, I will calculate the raw gap between the best and second-best competitor:

$$G = S_1 - S_2$$

where S_1 is the highest score and S_2 is the second highest score.

To standardize this gap against the wider field, I will compare it to the standard deviation of the full comparison field:

$$\gamma = z_{\text{field gap}} = \frac{G}{\sigma_{\text{field}}}$$

where γ measures how large the first-second gap is relative to the spread of the broader field.

Finally, I will standardize the same gap against the elite field:

$$\phi = z_{\text{elite gap}} = \frac{G}{\sigma_{\text{top 10}}}$$

where ϕ measures how large the first-second gap is relative to the spread of the top ten competitors.

Author note: the scary math symbols are there to make the vector representation cleaner :)

4.4 Z-Score Construction

This part of the paper gets quite subjective as I am the primary source deciding which metric defines the best. Omitting the Digital Popularity category, each domain has a measurable Win/Loss ratio for individual players AND/OR an elo system.

Keep in mind, statistically, even if an individual player/competitor has outlier scores, their WLR should average out to something accurate over time. Given that most of the players taken for this statistic are pros/players who have been around for a while, factors such as teamplay can be omitted in the average because they've been on multiple different teams throughout their career, and their impact is the factor I'm measuring.

5 Data Collection

5.1 osu!standard Data

For osu!standard, performance points (pp) were used as the primary metric. The broader field was defined as the top 100 players, while the elite field was defined as the top 10 players.

$$\alpha = z_{\text{avg gap}} = \frac{33570 - 22011.45}{3012.42} \approx 3.84$$

$$\beta = z_{\text{elite}} = \frac{33570 - 29279.7}{2497.48} \approx 1.72$$

$$\gamma = z_{\text{field gap}} = \frac{699}{3012.42} \approx 0.23$$

$$\phi = z_{\text{elite gap}} = \frac{699}{2497.48} \approx 0.28$$

$$\vec{d}_{\text{osu}} = \begin{bmatrix} \alpha \\ \beta \\ \gamma \\ \phi \end{bmatrix} = \begin{bmatrix} 3.84 \\ 1.72 \\ 0.23 \\ 0.28 \end{bmatrix}$$

Sourced from https://osu.ppy.sh/rankings/osu/global, using pp as the metric for data.